



Hackathon Project • Live Deployment Available

Smart Secure Student Attendance Management System



Multi-Factor Attendance Verification using
Face Detection, GPS Geo-Fencing & Dynamic Session Codes



Prevent
Proxy Attendance

Multi-layer verification
ensures genuine presence



Real-Time
Monitoring

Live attendance tracking
and instant analytics



AI-Assisted
Verification

Face recognition & location
validation for accuracy



Face Verification



GPS Geo-Fencing



Session Codes



Role-Based Access

#BuildTheFuture

Attendance Overview



Total Students

45

Present

38

Absent

7



Attendance Rate

Date	Subject	Present	Absent	Attendance %
20 May	AI Basics	38	7	84%
21 May	Data Science	40	5	89%
22 May	ML Fundamentals	37	8	82%
23 May	Cloud Computing	41	4	91%
24 May	Power Platform	39	6	87%



Submitted by
Khileshwar Sahu



Theme: AI at Work — Productivity & Teamwork



Contact: skhileshwar075@gmail.com



GitHub: github.com/skhileshwar075-cell/smart-student-attendance-system.git

Live Demo & Project Access



Scan to access the deployed
SmartAttend platform and explore
the complete working solution.

Team Introduction



Khileshwar Sahu

Solo Developer & Project Creator



Education

B.Sc. Graduate – Computer Science Enthusiast



Roles & Contributions

Full Stack Web Development (React, Node.js, PostgreSQL) | Android Development (Kotlin, MVVM) | System Architecture & Database Design | Security Implementation (JWT, Geo-fencing, RBAC) | AI Integration (TensorFlow.js, ML Kit, Anomaly Detection) | Testing, Deployment & Documentation



Project Development

Designed, developed, tested, and documented the complete **SmartAttend** system independently.



Tools & Approach

Leveraged AI-assisted development tools to accelerate implementation, debugging, and documentation while maintaining full ownership of architecture and engineering decisions.



GitHub Repository

github.com/skhileshwar075-cell/smart-student-attendance-system.git

Scan QR to view the repository



AI Tools Used

- ✓ GitHub Copilot
- ✓ ChatGPT (OpenAI)
- ✓ Microsoft Copilot
- ✓ Tabnine AI
- ✓ TensorFlow & ML Kit

Tech Stack

- React.js
- Node.js
- PostgreSQL
- Kotlin (Android)
- TensorFlow.js, ML Kit



Theme

AI at Work –
Productivity & Teamwork



Contact

skhileshwar075@gmail.com



Live Demo & Project Access
smart-student-attendance-system-ten.vercel.app/





Introduction



The **Student Attendance Management System** is a web-based and mobile application solution designed to track and manage attendance digitally. It aims to streamline the attendance process in educational institutions by improving accuracy, efficiency, and accessibility.



Digital attendance minimizes errors, saves time, and enhances the accuracy and efficiency of record-keeping.





Problem Statement

Problems with manual attendance:



Manual attendance consumes 10–15 minutes per class, Teachers spend excessive time taking attendance and following up



Manual processes cause delays, errors, and proxy attendance



Attendance data is fragmented and difficult to analyze



Students lack transparency and correction mechanisms



Institutions need secure, scalable attendance management



Difficult to track attendance record by teacher and students



Administrators lack timely analytics to identify low-attendance trends



These challenges lead to inefficiencies, inaccurate records, and increased administrative burden.

It's time to automate. It's time to transform.



Identify the Problem



Understand the Impact



Drive the Change

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Target Users & Impact



Primary: Teachers
Classroom attendance workflows



Secondary: Admins & Students
Admins (reporting) and Students (self-service corrections)



› **Reduce attendance workload** – Save valuable time for educators and administrators.



› **Prevent proxy attendance** – Minimize false/proxy attendance and ensure authenticity.



› **Improve reporting accuracy** – Generate accurate reports and insights effortlessly.



› **Enable real-time monitoring** – Track attendance in real-time across classes and departments.



› **Support data-driven academic decisions** – Leverage analytics to improve student outcomes.



› **Improve transparency & reduce errors** – Enhance accuracy and trust in attendance data.



› **Digitize and simplify recording** – Eliminate manual attendance marking through digital.

Attendance Overview

May 2024

Total Classes

124



Average Attendance

92% ↑ 8%

vs last month

Absent Students

18

↓ 6%

vs last month

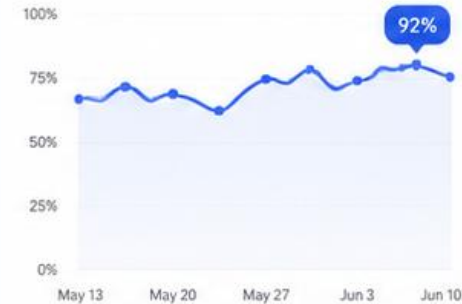
Reports Generated

32

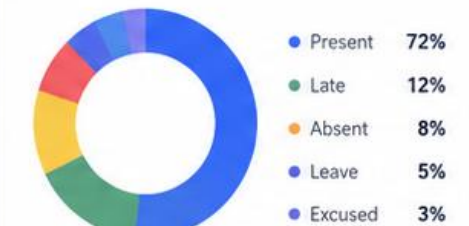
↑ 15%

vs last month

Attendance Trend



Attendance by Status



Transform attendance management.

Drive efficiency, accuracy, and trust across your institution.



Save Time



Increase Accuracy



Better Insights



Empower Everyone

#BuildTheFuture

Solution Overview



- Teacher launches a session with code/QR/secure mode + optional geofence



- Student marks attendance via web or Android client with **face + location** validation



- Backend **validates** sessions, prevents duplicates, logs anomalies, and updates analytics



- Real-time notifications** keep teachers, admins, and students aligned



- Dashboards** surface low-attendance risk, trends, and audit history



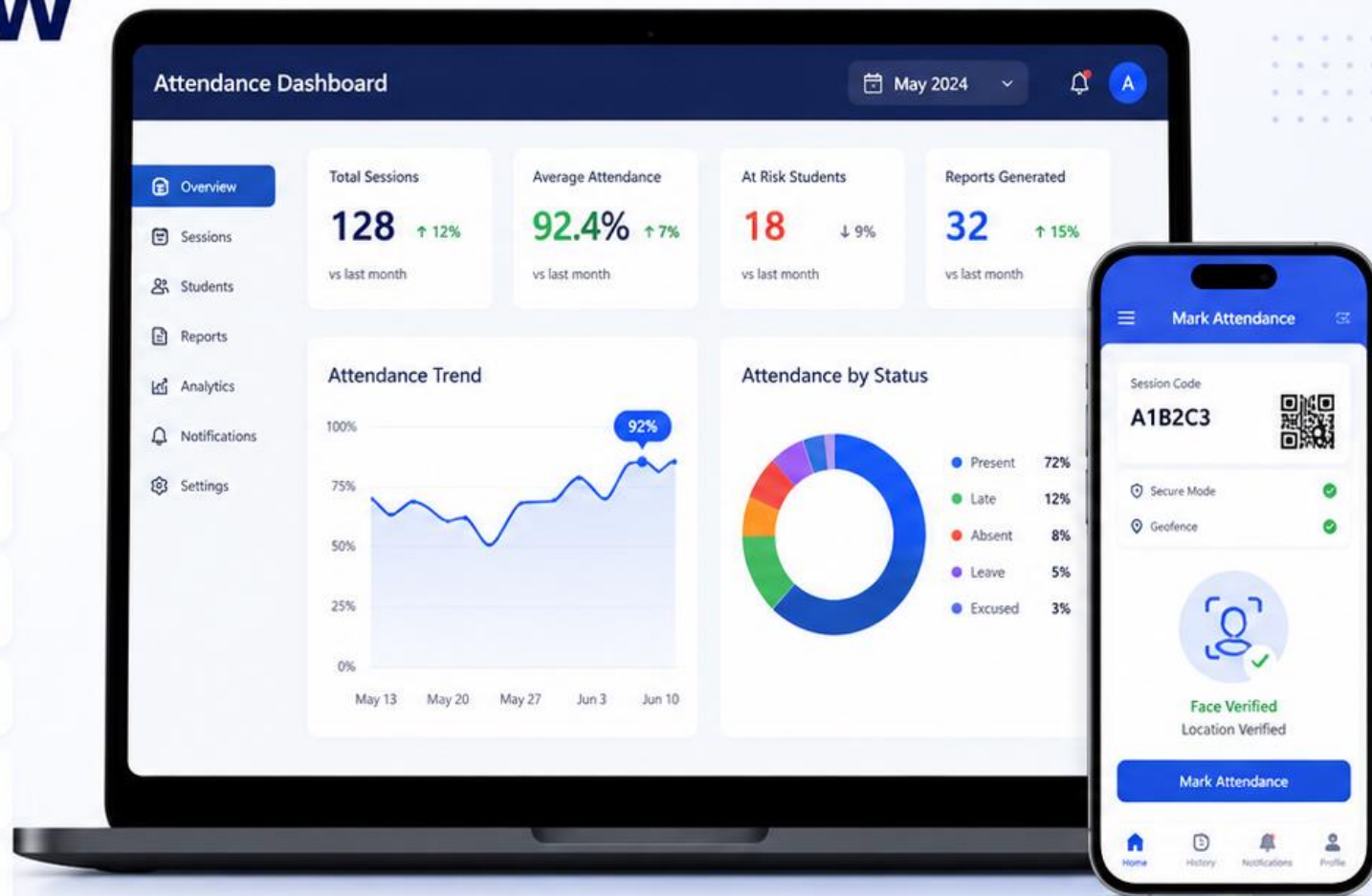
- One-click **Report Generation**



- Multi-platform **Web + Android** Access



- Correction Request **Workflow**



Accurate & Reliable



Secure & Trusted



Real-time & Efficient



Data-driven Insights

#BuildTheFuture

AI Integration

Leveraging AI and ML technologies to automate verification, detect anomalies, and deliver intelligent insights for smarter attendance management.



Current AI Features

- ✓ Face Verification using TensorFlow.js
- ✓ Android Face Detection using ML Kit
- ✓ Attendance Anomaly Detection
- ✓ Automated Low-Attendance Identification prioritized alerting



Future Enhancement

- ✓ Azure OpenAI Attendance Assistant
- ✓ Predictive Attendance Analytics
- ✓ Smart Recommendations & Insights
- ✓ Natural Language Query Support (Ask about attendance)



Intelligent
Face Verification



Anomaly
Detection



Smart
Alerts



AI-Powered
Insights

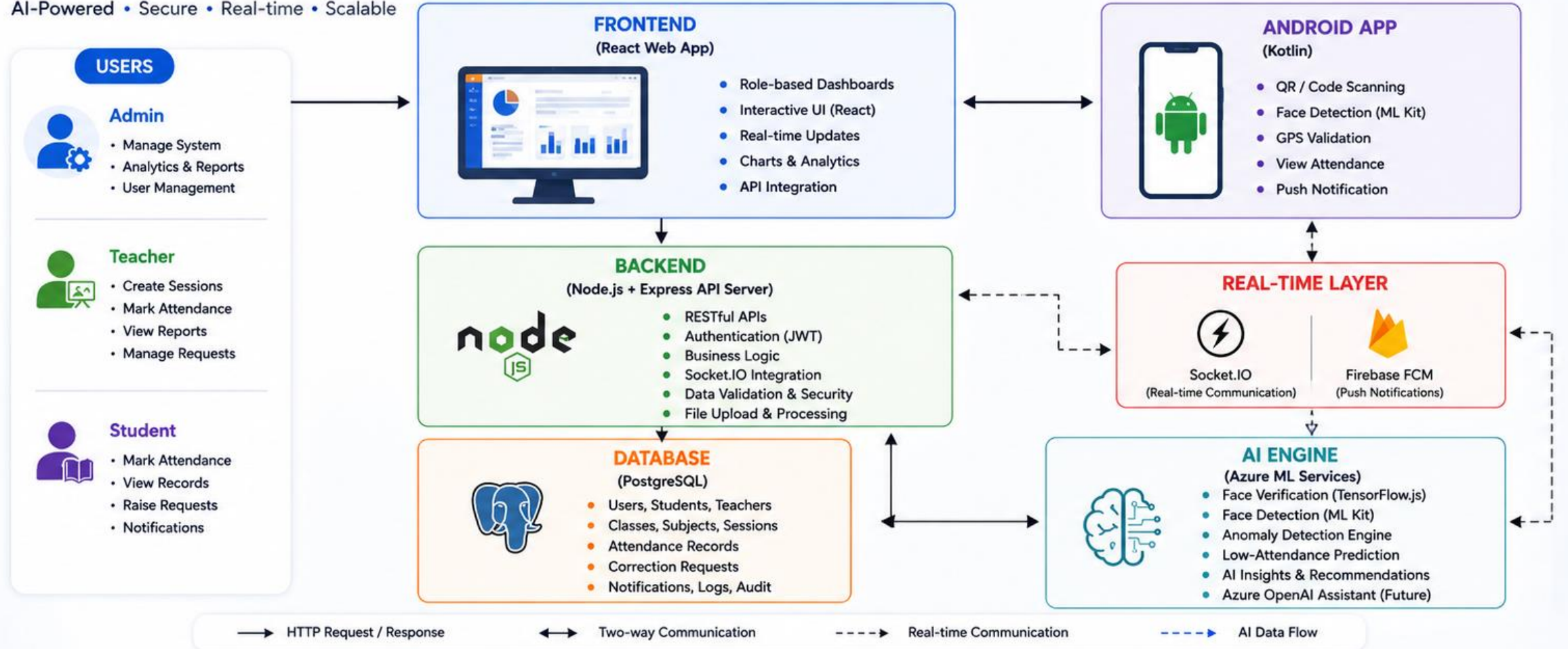


Cloud AI
Integration



Architecture Diagram

AI-Powered • Secure • Real-time • Scalable



KEY FEATURES



Secure
Authentication
(JWT)



AI Face
Verification



GPS
Geofencing



Real-time
Notifications



Analytics &
Dashboards



Audit Logs &
Anomaly Detection



Correction &
Approval Workflow

Live Demo Flow

- 01 Teacher creates attendance session
- 02 QR/Code generated
- 03 Student verifies face
- 04 GPS geofence validation
- 05 Attendance recorded securely
- 06 Analytics updated instantly
- 07 Notifications sent to stakeholders



Secure



Real-time



Accurate



Collaborative



Efficient

Teacher's Dashboard

×

Dashboard

Good afternoon, Dr.

Dr. Priya Sharma

Teacher

OVERVIEW

Dashboard

Students

Notifications

ATTENDANCE

Take Attendance

Active Sessions

Records

Student Requests

ANALYTICS

Reports

ACCOUNT

My Profile

Firebase • TensorFlow.js

Sign Out

Teacher Dashboard

Manage Your Classes

Wednesday, 03 June

3

My Subjects

7

My Students
unique, all classes

0

Present Today
unique students

7

Not Present
enrolled – present

Take Attendance

Start session or manual

View Reports

Analytics & exports

My Subjects

3 assigned

Activate Windows

Go to Settings to activate Windows.

Manual Attendance

×

Take Attendance

Good afternoon, Dr.



Dr. Priya Sharma

Teacher

OVERVIEW

Dashboard

Students

Notifications

ATTENDANCE

Take Attendance

Active Sessions

Records

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ANALYTICS

Reports

ACCOUNT

My Profile

Firebase + TensorFlow.js

Sign Out

Auto-refreshes every 15 seconds

Students (7)

All Present

All Absent

All Holiday

Mark Holiday

A	Aarav Singh	S001	✓ ✗ 📅
A	Ananya Gupta	S004	✓ ✗ 📅
K	Karan Mehta	S005	✓ ✗ 📅
K	Khileshwar Sahu	CD7286	✓ ✗ 📅
P	Priya Patel	S002	✓ ✗ 📅
R	Rahul Verma	S003	✓ ✗ 📅
S	Sneha Reddy	S006	✓ ✗ 📅

Present: 0

Absent: 7

Holiday: 0

Save Attendance

Activate Windows

Go to Settings to activate Windows.

Secure Attendance

✕ Take Attendance
Good afternoon, Dr.



Dr. Priya Sharma

• Teacher

OVERVIEW

Dashboard

Students

Notifications

ATTENDANCE

Take Attendance >

Active Sessions

Records

Student Requests

ANALYTICS

Reports

ACCOUNT

My Profile

Firebase • TensorFlow.js

Sign Out

SUBJECT

Data Structures & Algorithms — B.Tech CSE

DATE

05/30/2026

ATTENDANCE METHOD

Manual

Code

QR Code

Secure

Session Control

Geo-fence
Restrict to classroom location (100m)



Attendance Code

F809F1

Share with students • Active session running below

Face verification required

Secure Token

f075d3c73c06b7a5a57648cd803748ce

Rotate Token

Activate Windows
Go to Settings to activate Windows.

Attendance Report Generation

×

Reports

Good afternoon, Dr.

 **Dr. Priya Sharma**
Teacher

OVERVIEW

Dashboard

Students

Notifications

ATTENDANCE

Take Attendance

Active Sessions

Records

Student Requests

ANALYTICS

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My Profile

Firebase • TensorFlow.js

Sign Out

Attendance Report

Low Attendance

Data Structures & Algorithms

FROM

05/01/2026

TO

05/30/2026

Generate Report

Attendance Report

Summary

Detailed

Sneha Reddy

S006 • 8/14 classes

57.1%

Karan Mehta

S005 • 9/15 classes

60%

Ananya Gupta

S004 • 9/14 classes

64.3%

Rahul Verma

S003 • 9/14 classes

64.3%

Aarav Singh

S001 • 12/15 classes

80%

Priya Patel

S002 • 12/14 classes

85.7%

Activate Windows

Go to Settings to activate Windows.

Student Dashboard

Dashboard

Good afternoon, Aarav

Aarav Singh

Student

Dashboard

Mark Attendance

My Attendance

Requests

Notifications

Profile

Firebase • TensorFlow.js

Sign Out

OVERALL ATTENDANCE

88%

63 present · 9 absent · 0 holiday · 72 total

Attendance is satisfactory

Mark Attendance

Scan QR or enter code

Request Attendance

Submit correction

Subject-wise Attendance

4 subjects

Data Structures & Algorithms

15/19 classes attended

78.9%

Database Management Systems

16/19 classes attended

84.2%

Engineering Mathematics III

16/17 classes attended

94.1%

Operating Systems

16/17 classes attended

94.1%

Activate Windows

Go to Settings to activate Windows.

Student : Mark Attendance Page

Mark Attendance

Good afternoon, Aarav



Aarav Singh

Student

Dashboard

Mark Attendance

My Attendance

Requests

Notifications

Profile

Firebase • TensorFlow.js

Sign Out

Geolocation Status

FusedLocation API

Location captured

Lat: 20.66420, Lng: 81.55561 (±500m)

Enter Attendance Code

ENTER 6-DIGIT CODE

Submit

Scan QR

Face verification will be requested for secure attendance sessions

Active Sessions

Refresh

Data Structures & Algorithms

B.Tech CSE • Dr. Priya Sharma

Secure

3m 40s

Mark Present

Activate Windows

Go to Settings to activate Windows.

✕

My Attendance

Good afternoon, Aarav

Aarav Singh

Student

Dashboard

Mark Attendance

My Attendance

>

Requests

Notifications

Profile

🔗

Firebase • TensorFlow.js

➔

Sign Out

ATTENDANCE CALENDAR

May 2026

🗓

📅

Sun	Mon	Tue	Wed	Thu	Fri	Sat
					1 Absent	2 No record
3 No record	4 Absent	5 Absent	6 Absent	7 Present	8 Absent	9 No record
10 No record	11 Present	12 Present	13 Present	14 Absent	15 Present	16 No record
17 No record	18 Present	19 Present	20 No record	21 Present	22 No record	23 No record
24 No record	25 No record	26 No record	27 No record	28 Present	29 Present	30 No record
31 No record						

● Present days

● Holiday days

● Absent days

🔼 Filters

All Subjects

▼

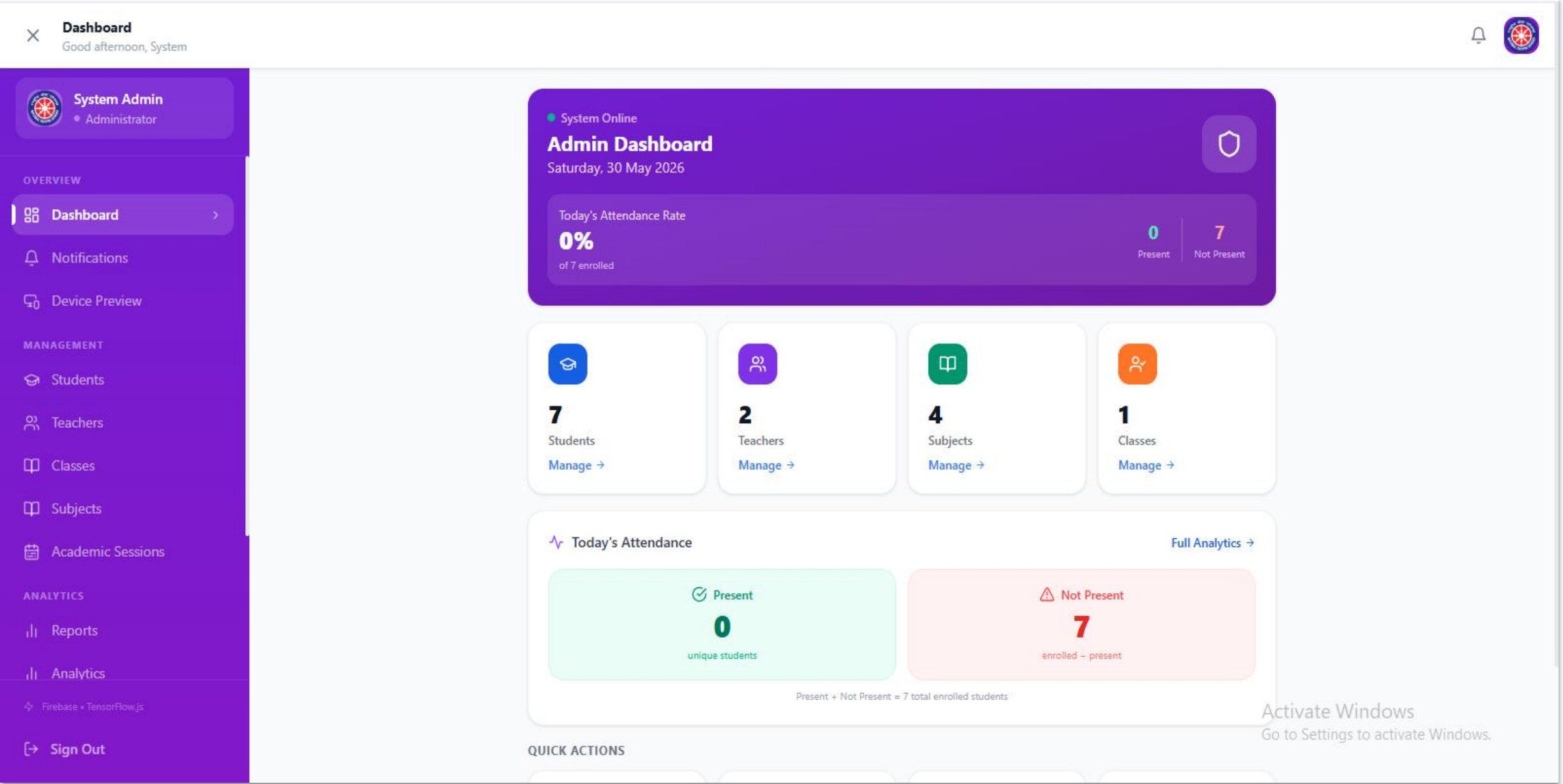
FROM DATE

TO DATE

Activate Windows

Go to Settings to activate Windows.

Admin Dashboard



Admin Analytics Page

